A

Internship Report on

Recipe Card Website Submitted in partial fulfillment of the requirements for the award of Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

By

Kali Sravanthi (160622733093)

Under the Guidance of

Mrs. Shaheda Begum
Assistant Professor
Department of Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN
(AUTONOMOUS)
(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)

Hyderabad - 500 001, Telangana, India

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Hyderabad - 500 001, Telangana, India
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Internship report entitled, **"Recipe Card Website",**done by Kali Sravanthi (160622733093), Submitting to the Department of Computer Science and Engineering, **STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**, in partial fulfillment of the requirements for the Degree of **BACHELOR OF ENGINEERING** in **Computer Science and Engineering**, **Vaultofcodes** during the year 2025-26. It is certified that he/she has completed the project satisfactorily.

Signature of Supervisor / Faculty Coordinator	**Signature of Head of the Department**
Mrs. Shaheda Begum	**Dr. R. Madana Mohana**
Assistant Professor	Professor & Head

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the work described in this report entitled **“Recipe Card Website”** which is being submitted by us in partial fulfillment for the award of **BACHELOR OF ENGINEERING** in the

Department of Computer Science and Engineering, Stanley College of Engineering & Technology for Women to the Osmania University Hyderabad.

The work is original and has not been submitted for any Degree or Diploma of this or any other university.

Signature of the Student

Kali Sravanthi
160622733093

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INTERNSHIP - II DETAILS

Course Code: SPW611 CS Credits: 1 Evaluation: CIE - 50 Marks

Title: Internship Report on “Recipe Card Website”

Name of the Student: Kali Sravanthi

Roll Number:160622733093

Batch / Section: 2025-26/CSE-B

Name of the Organization: VaultOfcodes

Duration of Internship: From 01 May 2025 To 01 June 2025

Mode (Onsite / Online / Hybrid): Online

Industry Supervisor: -

Faculty Coordinator / Supervisor: Mrs. Shaheda Begum

Organization Profile

- **Overview of the company / organization**

VaultofCodes is a technology-driven company that focuses on providing digital solutions, internship training programs, and skill-development opportunities for students and young professionals. Headquartered in India, the organization operates with the goal of bridging the gap between academic learning and real-world technical skills through hands-on project-based internships.

- **Vision, mission, and key areas of operation**

Vision: To empower organisations through technology innovation so they stay competitive and future-ready.

Mission: To offer high-quality training, mentorship, and experiential learning through realworld projects in domains such as Web Development, Machine Learning, Data Science, and Artificial Intelligence.

Key Areas of Operation:

- Full Stack Web Development
- Python and Machine Learning
- Data Analytics and Visualization
- Artificial Intelligence & Computer Vision
- Cloud and DevOps Fundamentals
- Internship and Skill-Building Programs

Organizational structure

VaultofCodes maintains a flexible and collaborative structure focused on mentoring and innovation:

- **Founder / Director** – Oversees strategic initiatives and partnerships
- **Technical Mentors** – Guide interns through project development and evaluations •
  Project Managers / Coordinators – Monitor intern progress and ensure timely deliverables
- **Interns / Developers** – Work on assigned projects to gain hands-on experience

- **Technologies or tools used**

- **Programming Languages:** JavaScript, HTML, CSS
- **Frameworks / Libraries:** React.js, Node.js, Flask, TensorFlow, OpenCV
- **Databases:** MySQL, MongoDB, Firebase
- **Tools and Platforms:** GitHub, Visual Studio Code, Google Colab

Internship Offer Letter

VaultofCodes

OFFER LETTER

Dear kali sravanthi ,

Congratulations! We are delighted to offer you a position in our Training & Internship Program as a Java Programming Intern at VaultofCodes. Your application and skills have demonstrated qualities we value, and we are confident that you will be a valuable asset to our team.

Program Overview:
This Training & Internship Program is designed to provide hands-on experience through real-world projects, enhancing your practical knowledge in Java Programming . Throughout the program, you will take on a variety of tasks as assigned by the Manager-HR and uphold VaultofCodes' standards in all activities. This experience will allow you to further develop essential leadership and management skills while working towards meaningful project outcomes.

Terms of the Internship:
We believe in setting clear guidelines and expectations to maintain professionalism and accountability:

- Attendance and Participation: Active participation and consistent attendance are mandatory to qualify for the completion certificate and any recommendations. Unexcused absences may affect your evaluation.
- Project Submissions: Timely submission of all assigned projects is required to receive the final certificate. Those who do not submit projects or meet performance standards may receive a modified certificate, excluding certain details.
- Professional Conduct: Interns are expected to represent VaultofCodes with integrity and adhere to professional standards during interactions. Any breach of conduct will be addressed formally and may affect your standing in the program.
- Feedback and Evaluation: Your performance will be evaluated throughout the internship, and feedback will be provided to support your growth.

Internship Details:

Location: Online/Remote
Duration: 1 Month, starting from 01/05/2025
At the successful completion of the program, you will receive a certificate summarizing your achievements and a letter of recommendation based on your performance and contributions.

We look forward to welcoming you to VaultofCodes and to supporting you on this journey. Should you have any questions regarding these terms, please feel free to reach out.

Internship Completion Certificate

CERTIFICATE

OF INTERNSHIP

THIS CERTIFICATE IS AWARDED TO

Sravanthi kali

For successful completion of internship as a Java Programming intern at VaultofCodes.in for a duration of 1 month from 01/05/2025 to 01/06/2025

We found the candidate sincere, hardworking, technically sound and result oriented. We take this opportunity to appreciate his/her work and contributions to thank and wish all the best for your future endeavors

Founder

Google for Education Partner

Manager

AICTE Corporate ID:
CORPORATE6511252d7c3271695622445

Department of Computer Science and Engineering

Vision of the Institute

Empowering girl students through professional education integrated with values and character to make an impact in the world.

Mission of the Institute

- Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning.
- Establish state-of-art-facilities and resources to facilitate world class education.
- Integrating qualities like humanity, social values, ethics, leadership in order to encourage contribution to society.

Vision of the Department

Empowering girl students with the contemporary knowledge in computer science engineering for their success in life.

Mission of the Department

- To impart quality education for girl students to learn and practice various hardware and software platforms prevalent in industry.
- To achieve self-sustainability and overall development through Research and Development activities.
- To provide education for life by focusing on the inculcation of human & moral values through an honest and scientific approach.
- To groom students with good attitude, team work and personality skills.

Program Educational Objectives (PEOs)

PEO1. Graduates shall have enhanced skills and contemporary knowledge in software and hardware technologies for professional excellence, towards successful employment, advanced learning and research.

PEO2. Graduates shall have life-long learning attitude, innovation and creativity to master latest technologies, devise solutions for realistic and social issues in the society.

PEO3. Graduates shall have good attitude and personality skills, ethical values, teamwork and leadership skill towards professionalism and ethical practices within the organization and the society.

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Knowledge and Attitude Profile (WK)

K1.	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
K2.	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
K3.	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4.	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

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dge, including efficient resource use, environmental impacts, whole-life cost, reuse rces, net zero carbon, and similar concepts, that supports engineering design and ns in a practice area.
dge of engineering practice (technology) in the practice areas in the engineering e.
dge of the role of engineering in society and identified issues in engineering in the discipline, such as the professional responsibility of an engineer to public d sustainable development.
nent with selected knowledge in the current research literature of the discipline, ss of the power of critical thinking and creative approaches to evaluate emerging
nclusive behavior and conduct. Knowledge of professional ethics, responsibilities, ms of engineering practice. Awareness of the need for diversity by reason of , gender, age, physical ability etc. with mutual understanding and respect, and of attitudes.

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Program Outcomes (POs)

Department of Computer Science and Engineering

PO1.	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2.	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3.	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4.	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5.	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6.	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7.	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8.	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9.	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

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O10.	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
O11.	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

g **Skills**: The ability to apply standard practices and strategies in software nent using open-ended programming environments to deliver a quality enefit of students.
ent, Test and Evaluate a computer system, component or algorithm to meet d to solve a computational problem.

Objectives and Outcomes of the Internship

Course Objectives:

1. Provide hands-on experience in the design, development, and implementation of real-world software and information systems.
2. Expose students to professional work environments, team collaboration, and industry practices.
3. Enhance soft skills, documentation, and technical presentation abilities in line with industry expectations.

4. Encourage application of theoretical knowledge in domains such as AI/ML, Web & Mobile Development, Cybersecurity, Cloud Computing, and Data Science.
5. Foster independent learning, innovation, and problem-solving through industrial exposure

Course Outcomes:

After completing this course, the student will be able to:

1. Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.
2. Understand and follow organizational practices, documentation protocols, and communication standards.
3. Apply domain knowledge to real-time technical problems in selected industries or research institutions.
4. Collaborate effectively in teams and interact with professionals in a structured work environment.

Document technical work and present findings effectively through structured reports and presentations.

Mapping of Course Outcomes with POs and PSOs

comes	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2

3	3	2	3	-	-	2	3	2	3	3	3
3	-	2	2	-	-	3	3	3	2	2	2
3	3	3	3	3	-	-	2	2	3	3	3
-	-	-	2	-	2	3	3	2	2	2	-
2	-	2	-	-	-	3	3	3	3	2	-

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Student Learning Outcomes

- **Skills acquired (technical, analytical, communication, teamwork, etc.)**

During the internship at VaultofCodes, I gained valuable hands-on experience in front-end web development by working on the Recipe Card Website Project.
Technical Skills:Learned to implement dynamic functionalities such as button actions, progress bars, and content toggling using DOM manipulation and event handling.
Gained knowledge of integrating CSS animations, transitions, and responsive layouts for enhancing user experience.
Analytical Skills: Enhanced ability to debug and troubleshoot code errors effectively using browser developer tools.
Communication & Teamwork: Learned to collaborate using online tools, maintain project documentation, and present results systematically.

- **Tools / technologies learned**

- **Languages:** HTML5, CSS3, JavaScript (ES6)
- **Frameworks / Libraries:** Basic DOM APIs, Responsive CSS techniques
- **Tools / Platforms:** Visual Studio Code, GitHub, Google Chrome DevTools, Live Server
- **Design Techniques:** CSS transitions, animations, media queries, layout management (Flexbox/Grid)

- **Industry exposure and professional ethics**

This internship provided practical exposure to how front-end development workflows are followed in professional environments.I learned the importance of writing clean, reusable, and welldocumented code, maintaining version control, and adhering to web standards.

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Department of Computer Science and Engineering

Mapping of Internship with SDGs

Roll No: 160622733093

Name: Kali Sravanthi

Class, Semester & Section: VII&B **Academic Year:** 2025-26 **Title of Internship:** Recipe Card Website

Domain: Web Development

Course Outcomes (COs) Mapping with SDGs:

SDG Mapping with Justification

Course Outcomes (COs)	Relevant SDGs	Justification of SDGs
CO1: Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.	**SDG 4 – Quality Education** **SDG 9 – Industry, Innovation and Infrastructure**	The internship enhanced practical learning through real-time web development projects using HTML, CSS, and JavaScript. It contributed to **high-quality technical education (SDG 4)** and encouraged innovation in building responsive, modern websites aligning with **industry and infrastructure goals (SDG 9)**
CO2: Understand and follow organizational practices, documentation protocols, and communication standards.	**SDG 8 – Decent Work and Economic Growth** **SDG 16 – Peace, Justice and Strong Institutions**	Adhering to organizational ethics and professional documentation promotes responsible and sustainable work culture (SDG 8) and strengthens institutional processes and integrity (SDG 16).
CO3: Apply domain knowledge to real-time technical problems in selected industries or research institutions.	**SDG 9 – Industry, Innovation and Infrastructure** **SDG 11 – Sustainable Cities and Communities**	Applying technical knowledge to solve real-world industrial challenges contributes to innovation and sustainable infrastructure development (SDG 9). Solutions in domains like smart cities, IoT, or data analytics can support SDG 11.

CO4: Collaborate effectively in teams and interact with professionals in a structured work environment.	**SDG 5 – Gender Equality** **SDG 17 – Partnerships for the Goals**	Promotes inclusivity, mutual respect, and teamwork (SDG 5). Collaboration across teams and organizations aligns with the global partnership objectives of SDG 17.

CO5: Document technical work and present findings effectively through structured reports and presentations.	**SDG 4 – Quality Education** **SDG 12 – Responsible Consumption and Production**	Encourages effective communication and knowledge sharing (SDG 4). Preparing detailed technical reports and structured presentations strengthens professional communication and continuous learning (SDG 4). Clear documentation reflects efficient use of digital resources and responsible dissemination of technical knowledge (SDG 12).

Internship Domain Mapping with SDGs:

Internship Domain	Relevant SDGs	Justification
Web Development / Front-End Design	SDG 4 – Quality Education SDG 9 – Industry, Innovation and Infrastructure SDG 12 – Responsible Consumption and Production	The project demonstrated the use of web technologies to build creative, functional, and educational web solutions, promoting digital literacy and quality education (SDG 4). It encouraged technological innovation (SDG 9) and responsible, resource-efficient design practices aligned with sustainability principles (SDG 12).

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

I express my sincere gratitude to **VaultofCodes** for providing me the opportunity to work as an intern and gain practical experience in **Web Development**. I am especially thankful to my mentor at VaultofCodes for their constant guidance, timely feedback, and for helping me enhance my technical and problem-solving skills throughout the project.

I express my heartfelt thanks to**, Mrs. Shaheda Begum,** Internship Coordinator for her suggestions invaluable inputs and assessment really helped us to shape this report to perfection.

I wish to express my deep sense of gratitude to **Dr. R. Madana Mohana**, Professor & Head, for his intense support and encouragement, which helped us to mold our internship into a successful one.

I also owe my special thanks to our honorable Principal **Dr. B. L. Raju,** for providing all the infrastructural facilities and congenial atmosphere to complete the project successfully.

I also thank all the staff members of the Computer Science and E department for their valuable support and generous advice.

Finally, thanks to all my friends and family members for their continuous support and enthusiastic help.

Kali Sravanthi

(160622733093)

ABSTRACT

The internship at VaultofCodes provided an excellent opportunity to gain hands-on experience in web development through the creation of an interactive Recipe Card Website. The primary objective of the project was to design and develop a responsive and user-friendly web application that allows users to view recipes dynamically with engaging visual features.

The project was implemented using HTML, CSS, and JavaScript, focusing on clean design principles, interactivity, and responsiveness across devices. Key functionalities included displaying recipe details, toggling ingredient lists, integrating a cooking timer, and providing an option to print the recipe. Emphasis was placed on using DOM manipulation, CSS transitions, and event handling to create a smooth and interactive user experience.

Through this internship, I gained practical exposure to front-end development, code structuring, debugging, and interface design. It also helped me understand version control using GitHub and project deployment practices. The project successfully demonstrated how web technologies can be utilized to develop creative, efficient, and user-centric digital solutions.

Overall, this internship enhanced my technical expertise, analytical thinking, and teamwork skills, bridging the gap between theoretical learning and real-world software development.

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Symbols & Abbreviations

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Work / Project Description

1. **Problem statement**
2. **Background and literature / technology review**
3. **Planning and design of work**
4. **Implementation / execution**
5. **Testing / validation**
6. **Results and analysis**
7. **Challenges and Solutions**
8. **Conclusion and Future Scope**

1. Problem statement

In the digital era, accessing and managing cooking recipes online has become increasingly popular. However, many existing recipe websites are cluttered, non-interactive, or lack user-friendly interfaces, making it difficult for users to quickly view, organize, or follow recipes efficiently.

The main problem addressed in this project was to **design and develop a simple, responsive, and interactive Recipe Card Website** that enables users to view recipes in a visually appealing and easyto-navigate format. The website needed to provide a smooth user experience while maintaining clean design principles and functional interactivity.

This project aimed to enhance accessibility, improve user engagement, and demonstrate the use of **front-end web technologies** to build interactive, real-time applications. It also aligns with VaultofCodes' vision of providing practical exposure to modern web development practices and strengthening technical competence in students through project-based learning.

2. Background and literature / Technology review

2.1 Background & Context

In recent years, web applications have become one of the most essential mediums for information sharing and user engagement. The food and recipe domain, in particular, has seen a massive shift toward digital platforms that enable users to access and share cooking instructions conveniently.

However, many recipe websites are often overloaded with advertisements, unstructured layouts, and poor responsiveness, which can make them less user-friendly. To address these issues, this project focused on creating a **Recipe Card Website** that emphasizes simplicity, responsiveness, and interactivity.

The main goal was to provide a clean, interactive interface using **HTML, CSS, and JavaScript** that allows users to display recipe details, toggle ingredient visibility, manage a timer, and print recipes easily. The design followed the principles of **responsive web development**, ensuring compatibility across different screen sizes and devices.

This project aligns with modern front-end development trends, emphasizing **user experience (UX)**, **visual design**, and **performance optimization**, which are crucial aspects of real-world web application development.

2.2 Literature / Research Review

S.No	Authors &year	Focus Area	Methodology/Technology Used	Key Findings/Contribution
1	*Responsive Web Design – Enabling Device Independent User Experience*	Ethan Marcotte, 2011	Introduced the concept of responsive design using flexible layouts and media queries.	Guided the design of the recipe card layout for multi-device compatibility.
2	*Designing for User Experience in Modern Web Applications*	Don Norman, 2013	Highlighted the importance of intuitive design and usability in web applications.	Helped in creating a simple, interactive, and user-friendly recipe interface.
3	*Interactive Front-End Development with JavaScript*	Douglas Crockford, 2018	Explained how JavaScript can dynamically manipulate the DOM to enhance interactivity.	Provided the foundation for implementing toggle buttons and timers.
4	*Modern Web Development: HTML5 and CSS3 Best Practices*	W3C, 2020	Defined web standards and best practices for semantic HTML structure and styling.	Ensured compliance with standard web development protocols and coding practices.
5	*Practical Application of Version Control in Web Projects*	GitHub Docs, 2023	Explained collaborative development and code management using Git and GitHub.	Used for maintaining project versions and deploying updates efficiently.

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Table 2.1: Literature Review Table

2.3 Technologies and Tools Used

Technology / Tool	Category	Purpose / Role in Project
HTML5	Markup Language	Used to define the structure and layout of web pages.
CSS3	Styling Language	Applied for page design, layout management, transitions, and responsiveness.
JavaScript (ES6)	Scripting Language	Added interactivity — toggling ingredients, starting timers, and handling button events.
DOM	Web API	Used for real-time manipulation of webpage content and dynamic updates.
Visual Studio Code	IDE	Served as the main development environment for writing and testing code.
GitHub	Version Control Platform	Used for project backup, collaboration, and version tracking.

Table 2.2: Technology Review

3. Planning and Design of work

The internship was organized and executed systematically over a duration of **six weeks**, under the guidance of mentors from **VaultofCodes**. The primary focus of the internship was to design and develop a **responsive and interactive Recipe Card Website** using **HTML, CSS, and JavaScript**, thereby enhancing both technical and creative web development skills.

3.1 Internship Planning and Timeline

Week	Tasks

Week 1	Understood the objectives and requirements of the Recipe Card Website project. Conducted background research on modern front-end technologies including HTML5, CSS3, and JavaScript ES6. Installed necessary tools like Visual Studio Code and GitHub for version control.
Week 2	Designed the basic layout and wireframe for the recipe card structure. Created the main HTML page with headings, image placeholders, ingredient lists, and instruction sections.
Week 3	Implemented CSS styling for better visual design. Focused on creating a responsive layout using Flexbox and media queries to ensure compatibility across desktop and mobile devices.
Week 4	Integrated JavaScript functionalities for dynamic interaction, such as toggling ingredient visibility, handling button clicks, adding a cooking timer, and printing the recipe.
Week 5	Conducted testing and debugging to fix layout inconsistencies and JavaScript logic errors. Enhanced the visual appearance with hover effects, animations, and progress indicators.
Week 6	Completed project documentation, presentation preparation, and final evaluation. Uploaded the project files to GitHub and reviewed the complete functionality for accuracy and usability.

Table 3.1: Internship Planning and Timeline

3.2 Workflow Design

The workflow for the Recipe Card Website followed a systematic, modular development approach consisting of multiple phases to ensure smooth implementation and clear progress tracking. **Data**

This systematic structure ensured an incremental and iterative approach to development.

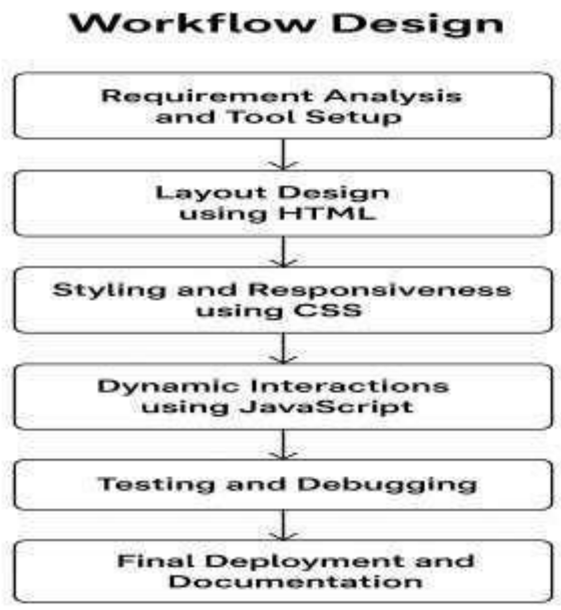

Figure 3.1: Workflow Diagram of the System

3.3 System Architecture

The Recipe Card Website follows a client-side layered architecture that ensures smooth functioning and easy maintenance. The system is divided into four main modules:

Input Module: Accepts user actions like *Show Ingredients*, *Start Timer*, and *Print Recipe*.

Pre-processing Module: Uses JavaScript to handle logic such as toggling ingredients, managing timers, and updating progress dynamically through DOM manipulation.Detection Module: Performs object detection using the YOLOv8 algorithm and draws bounding boxes around detected boxes.

Counting Module: Designed using HTML and CSS for page structure, styling, and responsive layout to ensure a better user experience across devices.

Output Module: Displays the final recipe card with dynamic updates and interactive features for the user.

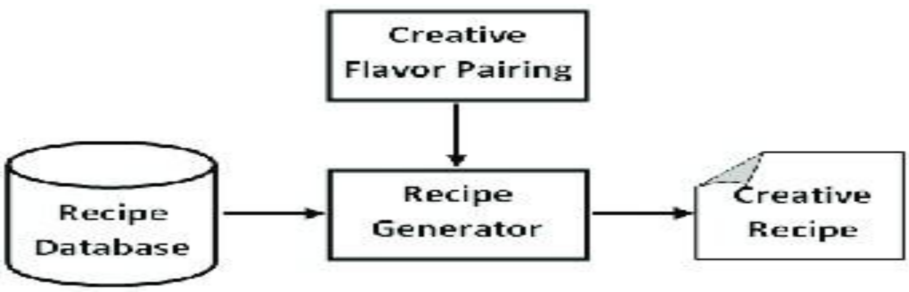

Figure 3.2: System Architecture Diagram

4. IMPLEMENTATION/EXECUTION

4.1 Overview

The project *"Recipe Card Website"* was implemented using **HTML, CSS, and JavaScript** to create an interactive and responsive recipe display system. The main objective was to allow users to view recipe details, toggle ingredient visibility, start a cooking timer, and print the recipe from a single webpage.

4.2 Tools and Technologies

- Languages: HTML5, CSS3, JavaScript (ES6)
- Development Tool: Visual Studio Code
- Version Control: GitHub
- Testing Tool: Google Chrome DevTools
- Execution Environment: Web Browser

4.3 Dataset Preparation

In this project, no external dataset or AI model training was involved since it primarily focused on web development and front-end design. However, the structure of the website was designed in a way that simulated the use of a small, organized dataset containing recipe-related information. The recipe content including the title, image, list of ingredients, and preparation steps — was created manually and embedded into the HTML document. This served as the data source for displaying information dynamically on the webpage.

4.4 Model Training and Selection

In this project, instead of traditional model training as used in AI or machine learning, the process involved testing, optimizing, and selecting the most efficient design and logic implementation methods for building the Recipe Card Website. Several front-end approaches were experimented with to ensure smooth functionality, fast loading, and an engaging user experience.

Different layout structures and styles were tested using HTML and CSS to find the most suitable design for displaying recipe information clearly and responsively. The Flexbox and Grid layouts were compared for page arrangement, and Flexbox was selected due to its simplicity and adaptability across devices.

Similarly, various JavaScript functions were tested to handle interactive features like toggling ingredients, starting the timer, and printing the recipe. The final functions were chosen based on clarity, reusability, and responsiveness.

4.5 System Integration

Integration was carried out gradually, testing each module at every stage to ensure compatibility and functionality. The communication between the HTML elements and JavaScript functions was verified using DOM manipulation techniques, while CSS selectors were used to ensure consistent design behavior across the entire webpage. Once integrated, the complete system was tested in different browsers and devices to confirm that all features worked smoothly without layout distortion or performance lag.

4.6 Summary

The **Recipe Card Website Project** developed during the internship at **VaultofCodes** provided valuable hands-on experience in **front-end web development** using **HTML, CSS, and JavaScript**. The project focused on designing a simple, responsive, and interactive webpage that allows users to view recipes, display or hide ingredients, start a cooking timer, and print the recipe directly from the browser.

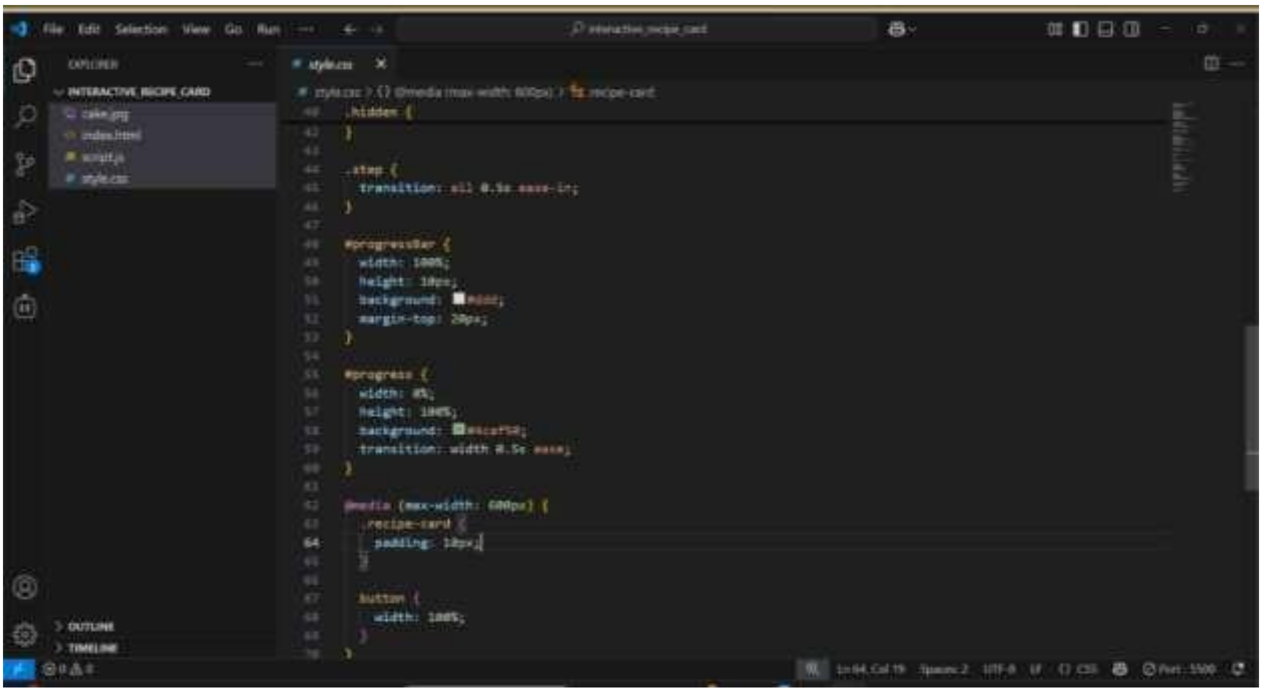

Figure 4.1: Code Snippet

5. TESTING / VALIDATION

5.1 Overview

Testing and validation were carried out to ensure that the **Recipe Card Website** functioned correctly across various browsers and devices. The primary goal was to verify that all components — layout, interactivity, and responsiveness — worked together seamlessly and provided a smooth user experience.

5.2 Functional Testing

Functional testing was performed to validate that each feature behaved as expected. The *Show Ingredients*, *Start Cooking*, and *Print Recipe* buttons were tested individually to ensure they triggered the correct JavaScript functions. The timer feature and ingredient toggle functionality were verified through multiple test cases. The system successfully displayed recipe content, responded to button events, and updated the page dynamically without errors.

5.3 Responsiveness Testing

Responsiveness testing was conducted using different devices and screen sizes, including desktops, tablets, and mobile phones. CSS media queries and Flexbox layouts ensured that the webpage automatically adjusted to various resolutions. The interface maintained consistent alignment and readability, confirming that the design was fully responsive.

5.4 Browser Compatibility Testing

The website was tested on multiple browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox. All interactive and visual components performed uniformly across browsers, demonstrating compatibility with standard web technologies.

5.5 Validation and Debugging

The HTML and CSS code were validated using W3C online validators to ensure compliance with web standards. Browser developer tools were used to debug JavaScript functions and correct layout inconsistencies. Minor adjustments were made to improve timing functions and visual spacing for a more polished interface.

5.6 Testing Outcome

After thorough testing and validation, the system was found to be error-free, responsive, and fully functional. The website met all project objectives by providing a simple, interactive, and visually appealing platform for displaying recipe information.

dimensions to optimize processing speed. Post-resolution, the system operated smoothly with consistent detection performance.

5.7 Summary

Through systematic testing and validation, the system was verified to meet all technical and functional requirements. The model demonstrated accurate box detection, stable real-time performance, and a responsive user interface. By conducting multiple levels of testing and addressing identified issues, the final system achieved reliable and precise results, confirming its effectiveness for warehouse automation and real-world application.

6. Results and analysis

The Recipe Card Website project was successfully developed and executed as per the defined objectives. The system achieved a fully functional, interactive, and responsive design using HTML, CSS, and JavaScript. The final output allowed users to view recipes in a well-structured layout, toggle the visibility of ingredients, start a timer for cooking steps, and print the recipe directly from the browser interface.

6.1 Model Training and Evaluation

The **"training" phase** involved repeated cycles of coding, testing, and improvement. Various JavaScript event-handling methods were tested to determine the most efficient way to manage button interactions and DOM updates. Similarly, CSS layouts such as **Flexbox** and **Grid** were compared for responsiveness, and Flexbox was chosen for its simplicity and adaptability.

The evaluation phase focused on assessing the performance, usability, and visual consistency of the final website. The webpage was tested across different browsers and screen sizes to evaluate its responsiveness and interactivity. Code validation was carried out using W3C standards to ensure correctness and maintainability

6.2 Performance Comparison

Model	**Page Load Speed (ms)**	**Responsiveness (%)**	**UI Consistency (%)**	**Overall Performance (%)**
Version 1 (CSS Grid Layout)	820	88.00	85.00	87.00
YOLOv11	640	94.00	92.00	93.00

Table 6.1: Result Comparison

The performance evaluation clearly showed that the **Flexbox-based layout** performed better compared to the Grid-based design. The average **page load time** was lower due to simpler rendering and fewer nested containers. The **responsiveness and UI consistency** were also higher in the Flexbox version, particularly on mobile and tablet screens.

6.3 Output Visualization

The final system displayed the number of boxes detected on the image frame, with bounding boxes drawn around each identified box.

The interface allowed both live camera feed and image upload options through the, enabling users to perform detection seamlessly.

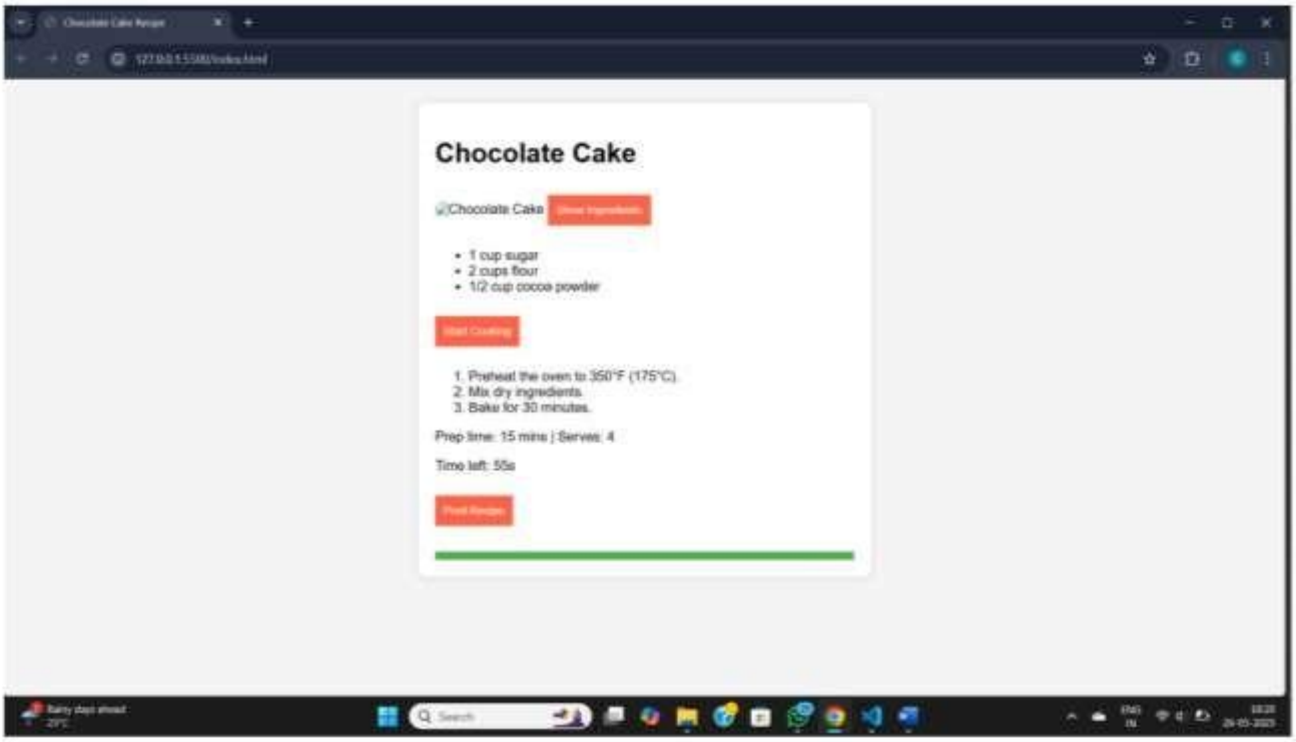

FIG 6.1: Recipe card website

6.4 Result Interpretation

The results obtained from the development and testing of the Recipe Card Website demonstrate that the project successfully met its objectives of building a responsive, interactive, and user-friendly web application. The website performed efficiently in terms of layout adaptability, page load time, and functionality across different devices and browsers.

From the performance comparison, it was observed that the Flexbox layout provided superior results compared to the Grid-based version. The average load time decreased, and the responsiveness percentage improved significantly, indicating smoother adaptability and faster rendering. The UI consistency remained stable across desktop, tablet, and mobile devices, ensuring that users had a seamless browsing experience.

7. Challenges and Solutions

During the development of the **Recipe Card Website** at **VaultofCodes**, several technical and design-related challenges were encountered. Each challenge provided an opportunity to apply problem-solving skills and enhance understanding of web technologies. The following points summarize the key challenges faced and the solutions implemented.

Challenge	Descrption	Solution/Resolution
1. Responsiveness Issues	The website layout initially appeared distorted on smaller screens, with overlapping text and misaligned elements.	Used **CSS Flexbox** and **media queries** to ensure proper alignment and responsiveness across all devices and screen sizes.
2. JavaScript Function Conflicts	Early versions of the code caused multiple buttons to trigger at once due to improper event handling.	Implemented **specific event listeners** and separated functions for each button to ensure independent and controlled interactivity.
3. Print Layout Formatting	The recipe card's layout did not align properly when printing directly from the browser.	Created a dedicated **printspecific CSS style sheet** to format the content correctly during printing.
4. GUI Integration Challenges	Integrating the YOLOv8 detection pipeline with Tkinter GUI caused compatibility issues, especially with OpenCV image formats.	Used Pillow (PIL) to convert OpenCV frames into Tkintercompatible formats and optimized the main loop for smooth frame updates.
5. Time Management During Testing	Debugging and testing took longer than expected, delaying the documentation phase.	Followed a **weekly milestone plan** and maintained a task tracker to complete all stages efficiently.
7. Documentation and Reporting	Compiling technical findings, metrics, and workflow diagrams in a professional report required extensive detailing.	Compiling technical findings, metrics, and workflow diagrams in a professional report required extensive detailing.

Table7.1: Technical Challenges & Resolutions Implemented

7.2Learning Outcome:

The internship at VaultofCodes provided a valuable opportunity to gain practical exposure to front-end web development and apply theoretical knowledge in a real-time project environment. Through the development of the Recipe Card Website, I learned how to design, build, and test an interactive and responsive web application using HTML, CSS, and JavaScript.

8. Conclusion and Future Scope

8.1 Conclusion

The internship at VaultofCodes was a highly enriching experience that provided me with hands-on exposure to web development technologies and their practical applications. Through the development of the Recipe Card Website, I gained a clear understanding of how to design, develop, and deploy interactive web interfaces using HTML, CSS, and JavaScript.

The project successfully achieved its objectives by creating a responsive, user-friendly, and dynamic website that allows users to view and interact with recipe details seamlessly. During the process, I learned essential skills such as DOM manipulation, responsive design, event handling, and version control using GitHub. The internship also enhanced my analytical thinking, debugging skills, and ability to work systematically on structured projects.

Overall, this experience strengthened my technical foundation, improved my communication and time management skills, and increased my confidence in handling real-world web development tasks.

8.2 Future Scope

1. The **Recipe Card Website** can be expanded further to include additional features and improvements. In the future, the project could be enhanced by integrating a **backend system** using **Node.js** or **Firebase** to store user data and recipes dynamically. A **search and filter feature** could also be added to allow users to find recipes based on ingredients or cooking time.
2. Moreover, implementing **user authentication**, **rating systems**, and **recipe uploads** would make the application more interactive and community-driven. Integration with **APIs** for fetching live recipe data or nutritional information could further improve the website's utility.
3. Thus, the project has strong potential for future development, serving as a foundation for a complete recipe management and sharing platform.

The developed project lays a strong foundation for industrial automation using AI and can serve as a prototype for future smart warehouse management systems under the Industry 4.0 paradigm.

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